SDC III (Android Application) Project Report

on

SmartCart

BACHELOR OF ENGINEERING

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

by

HEENA RATHORE (160723747060)

MANDA VAISHNAVI (160723747079)

KALA SANTHOSH (160723747104)

Under the Guidance of

Mrs. J. HARIKA

Assistant Professor, Dept. of CSE

Department of CSE (Artificial Intelligence and Data Science)

Methodist College of Engineering and Technology,
King Koti, Abids, Hyderabad-500001.

2025-2026

Methodist College of Engineering and Technology,
King Koti, Abids, Hyderabad-500001,

Department of Artificial Intelligence and Data Science

DECLARATION BY THE CANDIDATES

We, **Heena Rathore (160723748005), Manda Vaishnavi (160723748024), Kala Santhosh (160723748025)** students of Methodist College of Engineering and Technology, pursuing Bachelor's degree in Artificial Intelligence and Data Science , hereby declare that this Mini Project report entitled "**SMARTCART",** carried out under the guidance of **Mrs. J. Harika** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Artificial Intelligence and Data Science. This is a record work carried out by us and the results embodied in this report have not been reproduced/copied from any source.

Heena Rathore(160723747060)
Manda Vaishnavi(160723747079)
Kala Santhosh(160723747104)

Methodist College of Engineering and Technology,
King Koti, Abids, Hyderabad-500001.

Department of CSE (Artificial Intelligence and Data Science)

CERTIFICATE BY THE SUPERVISOR

This is to certify that this Mini Project work entitled "**SMARTCART**" *by* **Heena Rathore (160723747060), Manda Vaishnavi (160723747079), Kala Santhosh (160723747104)** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Artificial Intelligence and Machine Learning, during the academic year 2024-2025, is a bonafide record of work carried out by them.

Mrs. J. HARIKA.

Date: Assistant Professor, Dept. of CSE.

Methodist College of Engineering and Technology,
King Koti, Abids, Hyderabad-500001.

Department of CSE (Artificial Intelligence and Data Science)

CERTIFICATE BY HEAD OF THE DEPARTMENT

This is to certify that this Mini Project work entitled "**SMARTCART**" *by* **Heena Rathore (160723747060), Manda Vaishnavi (160723747079), Kala Santhosh(160723747104)** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Artificial Intelligence and Machine Learning, during the academic year 2024-2025, is a bonafide record of work carried out by them.

Dr. P. LAVANYA,
Professor & Head of the Department. CSE

Date:

Methodist College of Engineering and Technology,
King Koti, Abids, Hyderabad-500001.

Department of CSE (Artificial Intelligence and Data Science)

PROJECT APPROVAL CERTIFICATE

This is to certify that this Mini Project work entitled "**SMARTCART**" *by* **Heena Rathore (160723747060), Manda Vaishnavi (160723747079), Kala Santhosh(160723747104)** submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Artificial Intelligence and Machine Learning during the academic year 2024-2025, is a bonafide record of work carried out by them.

INTERNAL **EXTERNAL** **HOD**

ACKNOWLEDGEMENT

We would like to express a deep sense of gratitude towards the **Dr. Prabhu G. Benakop, Principal, Methodist College of Engineering and Technology,** for always being an inspiration and for always encouraging us in every possible way.

We would like to express a deep sense of gratitude towards the **Dr. Lakshmipathi Rao, Director, Methodist College of Engineering and Technology,** for always being an inspiration and for always encouraging us in every possible way.

Our sincere thanks to **Dr. P. Lavanya, Professor** and **Head of the Department of Computer Science and Engineering,** for her valuable guidance and encouragement which has played a major role in the completion of the project and for helping us by being an example of high vision and pushing towards greater limits of achievement.

We would like to express our sincere gratitude to my project guide **Mrs. J. HARIKA**, Assistant Professor, CSE, for giving us the opportunity to work on this topic. It would never be possible for us to take this project to this level without his innovative ideas and his relentless support and encouragement.

We are indebted to the Department of Computer Science & Engineering and Methodist College of Engineering and Technology for providing us with all the required facility to carry our work in a congenial environment. We extend our gratitude to the CSE Department staff for providing us to the needful time to time whenever requested.

We would like to thank our parents for allowing us to realize our potential, all the support they have provided us over the years was the greatest gift anyone has ever given us and also for teaching us the value of hard work and education. Our parents have offered us with tremendous support and encouragement, thanks to our parents for all the moral support and the amazing opportunities they have given us over the years.

METHODIST
COLLEGE OF ENGINEERING & TECHNOLOGY
[Autonomous Institution]
Accredited by NBA & NAAC with A+ Grade
Estd:2008 Approved by AICTE New-Delhi & Affiliated to Osmania University
COMPUTER SCIENCE & ENGINEERING DEPARTMENT

Vision & Mission

VISION

To become a leader in providing Computer Science & Engineering education with emphasis on knowledge and innovation.

MISSION

M1: To offer flexible programs of study with collaborations to suit industry needs

M2: To provide quality education and training through novel pedagogical practices

M3: To Expedite high performance of excellence in teaching, research and innovations.

M4: To impart moral, ethical valued education with social responsibility.

Program Educational Objectives

Graduates of Compute Science and Engineering at Methodist College of Engineering and Technology will be able to:

PEO1: Apply technical concepts, Analyze, synthesize data to Design and create novel products and solutions for the real-life problems.

PEO2: Apply the knowledge of Computer Science Engineering to pursue higher education with due consideration to environment and society.

PEO3: Promote collaborative learning and spirit of team work through multidisciplinary projects

PEO4: Engage in life-long learning and develop entrepreneurial skills.

Program Specific Outcomes

At the end of 4 years, Compute Science and Engineering graduates at MCET will be able to:

PSO1: Apply the knowledge of Computer Science and Engineering in various domains like networking and data mining to manage projects in multidisciplinary environments.

PSO2: Develop software applications with open-ended programming environments**.**

PSO3: Design and develop solutions by following standard software engineering principles and implement by using suitable programming languages and platforms

PROGRAM OUTCOMES

PO1: Engineering knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals, and an engineering specialization to develop the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society, and environment as required.

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

PO5: Engineering tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6: The engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national and international laws.

PO8: Individual and collaborative teamwork: Function effectively as an individual, and as a member or leader in diverse teams/multi-disciplinary teams.

PO 9: Communication: Communicate effectively and inclusively within engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, considering cultural, language, and learning differences.

PO10: Project management and finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-long learning: Recognize the need for and have the preparation and ability for i). independent and life-long learning ii). Adaptability to new and emerging technologies and iii). Critical thinking in the broadest context of technological change.

TABLE OF CONTENTS

S. No.	Title	Page No.
	Certificate	I
	Acknowledgement	II
	Abstract	III
1	Introduction	1
1.1	Objective	2
1.2	Scope	2
2	Literature Survey	3
3	System Analysis	5
3.1	Existing System	5
3.1.1	Drawbacks	5
3.2	Proposed System	6
3.2.1	Advantages	7
3.3	Applications	7
3.4	Technical Specifications	8
3.4.1	Software Requirements	8
3.4.2	Hardware Requirements	8
4	System Design	9
4.1	Block Diagram / Circuit Diagram	9
4.2	Flowchart	11
5	Implementation	13
6	Testing	15
7	Output Screens	18
	Conclusion	22
	Future Enhancements	23
	References	24
	Appendix (QR CODE with all project docs in Gdrive)	25

ABSTRACT

The rapid expansion of e-commerce platforms such as Amazon, Flipkart, and Meesho has provided consumers with a wide variety of options for purchasing products online. However, this variety also poses a challenge: identifying the platform that offers the lowest price and best quality for the same product. Traditionally, customers are required to visit multiple websites, compare prices manually, and evaluate reviews, which is both time-consuming and inefficient. This problem highlights the growing need for an automated solution that can provide accurate, real-time price comparisons across multiple platforms in a user-friendly manner.

SmartCart is proposed as an Android-based application designed to simplify and optimize the online shopping experience. The application allows users to search for products either through text-based queries or by uploading screenshots and photos of items, utilizing OCR and image recognition technologies. Once the product is identified, SmartCart compares its price, availability, and ratings across multiple e-commerce platforms and presents the results in a structured, easy-to-understand interface. This not only helps users save money and time but also enhances decision-making by providing product quality insights. The inclusion of advanced features such as image-based search and real-time data integration positions *SmartCart* as a modern, AI-assisted solution for today's digital shopping landscape.

Keywords— Price comparison, e-commerce platforms, Product Search, Lowest Price Detection, Cost-Effective Shopping.

1. INTRODUCTION

The rise of e-commerce platforms such as Amazon, Flipkart, and Meesho has transformed the way consumers shop by offering convenience, variety, and competitive pricing. However, the presence of multiple platforms also makes it difficult for users to determine which website provides the best price and quality for the same product. Manually visiting different sites, comparing prices, and checking ratings or reviews is often time-consuming and overwhelming. As online shopping continues to grow, there is a clear need for an intelligent tool that can simplify this comparison process and assist users in making informed purchasing decisions.

SmartCart is developed as an Android-based application that addresses this challenge by offering automated, real-time price comparison in a user-friendly interface. The application allows users to search for products either by typing the product name or by uploading a screenshot or photo of the item. Leveraging OCR and image recognition technologies, SmartCart identifies the product accurately and retrieves its price, rating, and availability across multiple e-commerce platforms. The results are then presented in a structured format with prices sorted in ascending order, enabling users to quickly identify the best deal.

With its combination of text-based search, image-based search, and platform redirection, SmartCart not only enhances the shopping experience but also ensures efficiency, cost savings, and better decision-making. The integration of modern technologies positions SmartCart as a smart, practical, and essential solution for today's digital shoppers.

I.1 Objectives of Smartcart:

1. To develop an Android-based price comparison application named SmartCart.
2. To help users find the lowest price and best-rated products across multiple e-commerce platforms.
3. To support both text-based and image-based product searches for user convenience.
4. To reduce the time and effort spent on manually comparing prices on different websites.
5. To provide a simple, intuitive, and responsive interface for a seamless shopping experience.
6. To enable direct redirection to the selected e-commerce website for quick purchasing.
7. To demonstrate the use of modern mobile technologies and intelligent automation in online shopping.

I.2 Scope:

The scope of **SmartCart** extends to all users who engage in online shopping and wish to compare prices efficiently without manually visiting multiple platforms. The system can be used by individuals seeking cost-effective purchasing options and businesses analyzing price trends. Its functionality is designed to be scalable, allowing integration with additional e-commerce platforms and features like live API connections, user accounts, and personalized product recommendations in the future. SmartCart demonstrates the potential of combining mobile technology, image recognition, and intelligent data comparison to create a smart, user-friendly, and practical shopping assistant for the modern digital marketplace.

2. LITERATURE SURVEY

S. No.	Title of Paper	Author & Year	Methodology / Technology Used	Advantages	Limitationns	Conclusion
1	A Comparative Study of E-Commerce Price Aggregators	R. Kumar, S. Patel (2020)	Web scraping, data integration, and aggregation algorithms	Helps compare product prices across websites	Limited to few e-commerce sites	Demonstrated how automated comparison improves decision-making
2	Price Comparison Application for Online Marketplaces	**A. Sharma, K. Joshi (2021)**	REST APIs, data crawling, mobile app interface	Real-time data access and faster comparison	Lacks image-based search support	Improved accuracy and usability for online shoppers
3	AI-Based Product Identification Using Image Recognition	M. Singh, P. Verma (2022)	CNN and TensorFlow for image classification	Accurate product detection using AI	High computational cost	Validated image recognition for smart shopping systems
4	OCR-Driven Mobile Applications for Smart Shopping	**S. Roy, D. Das (2021)**	OCR (Tesseract), Android Studio, text extraction	Enables recognition from screenshots	Struggles with poor image quality	Effective for quick product identification from images
5	Design and Development of a mobile shopping assistant	L. George, I. Mathew (2019)	Android app using Firebase backend	Easy product navigation and storage	No comparison feature	Created a base for personalized shopping assistance
6	Enhancing User Experience in E-Commerce Through Mobile Apps	K. Desai, P. Nair (2020)	Flutter framework, UI/UX optimization	Improved cross-platform design	Focused on UI only, not backend logic	Proved Flutter's efficiency for mobile shopping systems
7	Integration of Flutter and Firebase for Cross-Platform	P. Jain, R. Mehta (2022)	Flutter, Firebase Firestore, cloud-based data handling	Real-time synchronizatio n and scalability	Needs continuous internet connectivity	Demonstrated lightweight yet powerful app architecture

	Mobile Applications					
8	SmartCart – An AI-Assisted Price Comparison Application	Proposed System (2025)	Flutter, Firebase, OCR, Custom APIs	Dual search (text/image), price comparison, direct redirection	APIs are currently limited	Provides a complete, scalable, and user-friendly shopping solution

Table 2.1 – Literature survey

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Traditional price comparison methods require users to **manually visit multiple e-commerce websites**, search for the same product, and compare prices, ratings, and reviews individually. While some web-based price comparison tools exist, they often have **limited platform coverage**, lack **real-time updates**, and do not offer **mobile-friendly interfaces**.

Existing systems also **do not support image-based product searches**, which makes it difficult for users who have only a screenshot or photo of a product. Furthermore, many comparison tools are web-only, meaning they lack accessibility and convenience for mobile users who prefer app-based browsing.

1. **2019:** Basic price comparison websites list prices from few stores, lacking live updates and user personalization.
2. **2020:** Comparison extensions for browsers improve accessibility but are desktop-only and not optimized for mobile users.
3. **2021:** Mobile comparison apps introduced limited filtering options and no AI or image recognition.
4. **2022:** Some AI-based projects used keyword matching for comparisons but lacked real-time synchronization with platforms.
5. **2023:** Multi-platform comparison websites began integrating reviews but still missed camera/upload-based product identification.

3.1.1 DRAWBACKS OF EXISTING SYSTEMS

1. Require users to manually visit multiple e-commerce websites to compare prices.
2. Lack real-time updates, leading to outdated or inaccurate price information.
3. Most systems are web-based and not optimized for mobile devices.

4. Limited platform coverage, supporting only a few major e-commerce sites.
5. Absence of image-based search, restricting users who have product photos or screenshots.
6. Complex user interfaces that are not beginner-friendly.
7. Some systems prioritize affiliate marketing instead of unbiased user benefit.
8. Do not provide direct navigation to product pages on respective platforms.
9. Often experience slow data retrieval and poor synchronization between sources.

3.2 PROPOSED SYSTEM

The proposed system, **SmartCart**, is an Android-based price comparison application that provides users with an automated and efficient way to compare product prices across multiple e-commerce platforms. The system allows users to search for products through text input or image upload, enabling both traditional and visual search experiences.

The workflow of SmartCart begins with the initialization of the application, where users are presented with options to either type the product name or upload/capture a product image. Once an input is provided, the system processes the request using integrated logic to identify the product. For image-based search, the application utilizes image recognition techniques to extract the product details from the captured image.

After identification, SmartCart retrieves real-time or stored data from multiple platforms, including product name, price, and rating. The comparison logic then sorts the results in ascending order based on price, ensuring that the user can easily find the lowest available option. The interface also provides direct redirection links, allowing users to visit the respective e-commerce site to view or purchase the product.

The home screen additionally includes sections such as popular product categories and platform shortcuts, making navigation simple and efficient. This continuous cycle of user input, data processing, and result display ensures a seamless, interactive, and intelligent price comparison experience. The design aims to reduce manual effort, save time, and deliver accurate results.

3.2.1 ADVANTAGES OF PROPOSED SYSTEM

1. Provides automated and accurate price comparison across multiple e-commerce platforms.
2. Supports dual search functionality — both text-based and image-based product searches.
3. Ensures faster and more efficient results compared to manual price checking.
4. Displays product ratings, reviews, and prices together in one unified interface.
5. Offers direct redirection to the selected e-commerce website for easy purchasing.
6. Features a mobile-friendly and user-centric design built using Flutter.
7. Reduces user effort and time by eliminating the need to browse multiple platforms.
8. Allows easy scalability for integrating more platforms and advanced features in the future.
9. Incorporates error handling and validation, ensuring smooth app performance.
10. Enhances the overall shopping experience with an intuitive and responsive interface.

3.3 APPLICATIONS

1. Can be used by online shoppers to compare product prices across multiple e-commerce platforms easily.
2. Helps budget-conscious users find the best deals and save money while shopping online.
3. Useful for market analysts and retailers to study price trends and competitor pricing.
4. Can be implemented by e-commerce startups to enhance user engagement through integrated price comparison features.
5. Suitable for students and researchers working on projects related to data comparison, mobile applications, and e-commerce systems.
6. Can be extended into a browser extension or web application for wider accessibility.
7. Acts as a practical mobile tool for users who prefer convenience and quick decision-making during online shopping.

3.4 TECHNICAL SPECIFICATION

3.4.1 HARDWARE REQUIREMENTS

1. Android smartphone with Android 7.0 (Nougat) or higher.
2. Minimum 2 GB RAM for smooth operation.
3. Minimum 50 MB of free storage space.
4. Quad-core processor or above.
5. Internet connectivity (Wi-Fi or mobile data).

3.4.2 SOFTWARE REQUIREMENTS

1. Frontend Framework: Flutter (Dart SDK).
2. IDE / Development Tool: Android Studio.
3. Backend / Database: Firebase Firestore.
4. APIs Used: Custom APIs for data fetching and price comparison.
5. Libraries / Tools: Image recognition and OCR libraries for photo-based search.
6. Operating System: Windows 10 / 11 or macOS (for development).
7. Control: GitHub (optional for project management).

4. SYSTEM DESIGN

4.1 BLOCK DIAGRAM

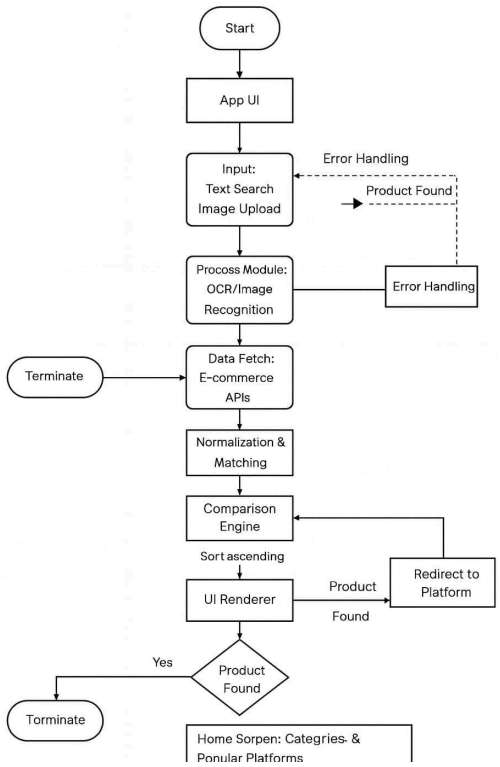


Fig 4.1 BLOCK DIAGRAM OF SMARTCART

Step 1: Start: The system begins when the user launches the SmartCart app on their Android device.

Step 2: App UI : The user interacts with the main user interface, which includes options like searching for products or uploading an image.

Step 3: Input (Text Search / Image Upload)

The user can either:

- Type the product name in the search bar, or
- Upload or capture an image of the product.

Step 4: Process Module (OCR / Image Recognition)- If the user uploads an image, the OCR and image recognition module processes it to extract text or identify the product visually.

Step 5: Data Fetch (E-commerce APIs)- Once the product is identified, the system retrieves product details, prices, and ratings from various e-commerce platforms using predefined custom APIs or stored data.

Step 6: Normalization & Matching- Fetched data is cleaned and standardized — product titles, prices, and ratings are normalized to ensure uniform comparison.

Step 7: Comparison Engine- This engine compares prices and ratings across all available platforms and sorts the results in ascending order (lowest price first).

Step 8: UI Renderer- The sorted comparison results are then displayed to the user through a well-structured interface, showing platform names, prices, and ratings.

Step 9: Redirect to Platform- When the user selects a particular product listing, the app redirects them to that e-commerce platform's webpage for purchase.

Step 10: Error Handling- If any issue occurs (like no product found, connection errors, or invalid input), the system triggers the **error** handling module to notify the user and allow re-entry.

Step 11: Home Screen (Categories & Platforms)- Users can also explore categories and popular platforms directly from the home screen for convenience.

Step 12: Terminate- After completing the comparison or redirection, the process terminates.

4.2 Flow Chart of SmartCart

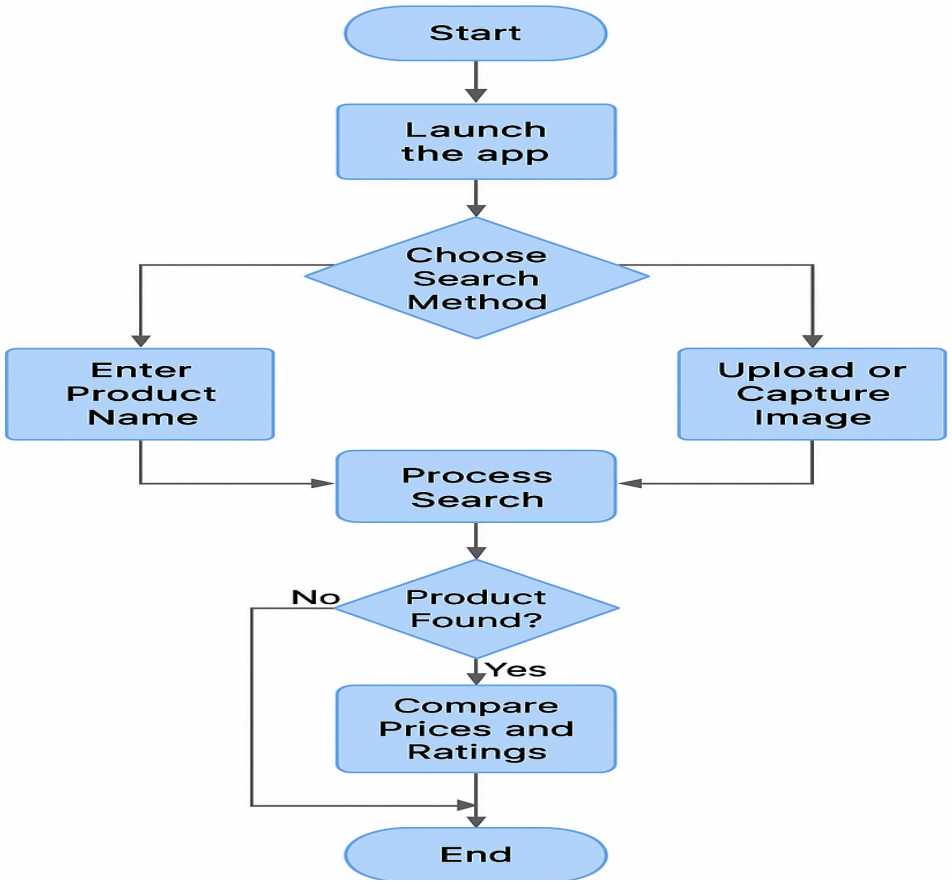


Fig 4.4 – FLOW CHART OF SMARTCART

Step 1: Start - The process begins when the user opens the SmartCart application.

Step 2: Launch the App - The system initializes the home screen and loads all necessary modules.

Step 3: Choose Search Method - The user selects the preferred search option — either text-based or image-based.

Step 4: Enter Product Name / Upload or Capture Image

- If the user chooses text search → they enter the product name manually.
- If the user chooses image search → they upload or capture a product image.

Step 5: Process Search - The application processes the input:

- For text: fetches relevant product data using APIs.
- For image: uses image recognition (OCR) to identify the product and retrieve details.

Step 6: Product Found? - The system checks if the product is available in the database or fetched data.

- If **No**, the system displays a message and allows the user to retry.
- If **Yes**, it proceeds to the next step.

Step 7: Compare Prices and Ratings - The comparison engine analyzes prices and ratings from multiple platforms, displaying the results in ascending order of price.

Step 8: End - The process ends once the results are displayed or the user exits the application.

5. IMPLEMENTATION

SmartCart – Intelligent Price Comparison Application is an Android-based system developed using **Flutter** in **Android Studio**. It is designed to provide users with an easy and efficient way to compare product prices across multiple e-commerce platforms. The system integrates technologies such as **Firebase**, **custom APIs**, and **image recognition** to deliver both text-based and image-based search functionality. Users can either type a product name or upload a photo, after which the app processes the input, retrieves prices and ratings, and displays results in ascending order to highlight the lowest price.

The implementation follows a modular design, ensuring that each component performs a distinct role while maintaining smooth data flow between the frontend and backend. The **frontend (Flutter UI)** handles user input and result presentation, while the **backend (Firebase and APIs)** manages data processing, retrieval, and comparison. This modular approach enhances maintainability, scalability, and performance, ensuring the app delivers an accurate and user-friendly experience.

5.1 MODULE DESCRIPTION

- Product Search (Text & Image)
- Data Processing
- Comparison Engine
- Display & Redirection
- Home Screen (Categories & Popular Platforms)

1. **Product Search Module**
   - Allows users to search for products either by typing the product name or uploading/capturing an image.
   - Uses keyword matching for text searches and image recognition logic for image-based searches.
   - Ensures flexible and efficient product identification to enhance user convenience.

2. **Data Processing Module**
   - Handles extraction and normalization of product data received from custom APIs and Firebase.
   - Organizes product information such as name, price, and rating into a uniform format.
   - Ensures accurate data handling and prepares it for comparison.

3. **Comparison Engine Module**
   - Compares product data from multiple sources and sorts it in ascending order of price.
   - Highlights the lowest price to help users identify the best deals quickly.
   - Maintains consistency and reliability in comparison results.

4. **Display and Redirection Module**
   - Displays comparison results with product name, rating, price, and platform name.
   - Provides direct redirection links to respective e-commerce platforms for product purchase.
   - Offers a smooth transition between result view and external platform access.

5. **Home Screen Module**
   - Provides categorized product sections for easy exploration.
   - Includes a list of popular platforms for direct navigation.
   - Ensures user-friendly design and quick access to essential features.

6. TESTING

6.1 IMPORTANCE OF TESTING

Testing is a crucial phase in the software development life cycle that ensures the application performs as intended, meets user requirements, and maintains reliability under various conditions. It helps identify errors, defects, and inconsistencies within the system before deployment, thereby improving software quality and user satisfaction.

In the case of **SmartCart**, testing plays a vital role in validating the functionality of all modules, including product search, data processing, comparison engine, and user interface operations. Since the app integrates multiple features such as text-based search, image recognition, and platform redirection, thorough testing is essential to ensure smooth interaction between these modules.

Moreover, testing verifies that the application retrieves and compares data accurately, displays consistent results, and provides a seamless experience across different Android devices. It also ensures the app handles errors efficiently—such as invalid inputs, failed data fetches, or network interruptions—without affecting usability. By performing rigorous testing, SmartCart achieves higher stability, efficiency, and performance, ultimately ensuring a reliable and user-friendly price comparison platform.

6.2 TESTING DETAILS

The testing of **SmartCart** was conducted methodically to ensure each component performed as expected and the system delivered accurate results.

The table provides a detailed overview of the primary test cases conducted during the testing phase of the **SmartCart** application. Each test case represents an essential functionality within the system, ensuring that all core modules perform as expected. For instance, the tests for product search through text input and image upload confirm that the app can accurately process different types of user queries and retrieve corresponding results from various e-commerce platforms. Similarly, the navigation test validates that the application correctly redirects users to the respective product pages on external websites without any errors

Test No.	Test Case	Expected output	Actual output	Result
1	Search product by name	Display products details with prices	Displayed correctly	pass
2	Upload image for search	Identify and show related products	Successfully detects product	Pass
3	Compare prices	Show results in ascending order	Display sorted list	Pass
4	Redirect to platform	Open respective product page	Redirected successfully	Pass
5	Handle invalid input	Display error message	Error handled smoothly	Pass

Table 6.2 – Testing details

The home screen display test ensures that all categories, popular platforms, and essential features load properly, contributing to a smooth user experience. The outcomes from each test indicate that both the expected and actual results are consistent, confirming that SmartCart functions efficiently under multiple scenarios. Overall, the testing process verifies the system's reliability, responsiveness, and accuracy, demonstrating that the application is well-optimized and ready for real-world use.

6.3 TEST CASES

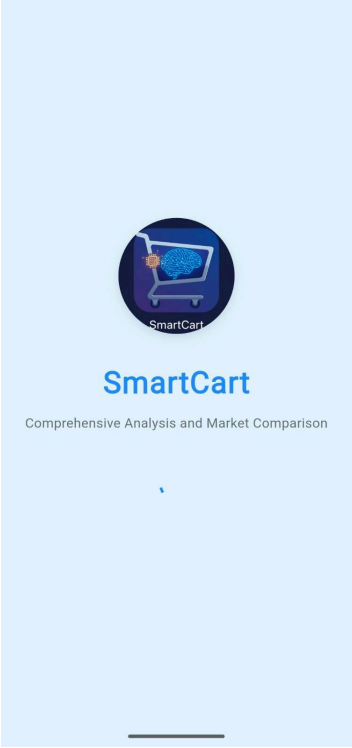
Test Cases

Test No.	Test Case	Description	Expected Input	Expected Output
1	Search Product	Testing the product search functionality through text input	Input product name	Lists prices and ratings from multiple platforms
2	Screenshot Search	Validating functionality of searching using screenshots	Upload screenshot of product	Displays prices and ratings for identified product
3	Image Search	Verifying search capability through photo upload	Upload photo of product	Shows prices and ratings for identified product
4	Category Selection	Checking navigation and selection of product categories	Select a category	Displays products and pricing within the chosen category
5	Platform Navigation	Testing direct access to e-commerce platforms	Tap on a platform link	Opens the selected platform in a web browser

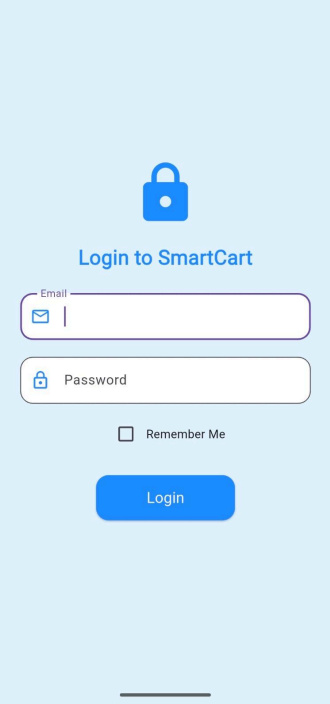
Table 6.3 – Test cases

7. OUTPUT SCREENS

7.1 - splash screen: SmartCart Launch Page
Displays the app logo and tagline: *"Comprehensive Analysis and Market Comparison."*

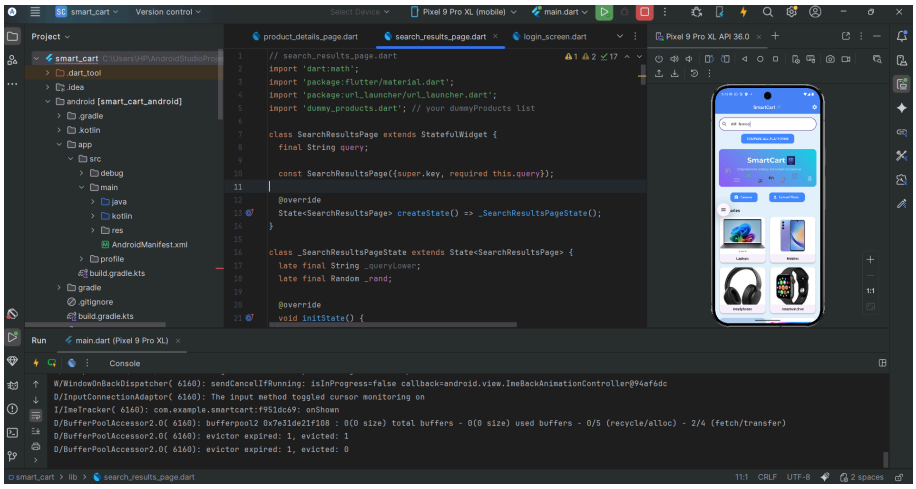


7.2 - Login Screen – User Authentication Interface, Allows users to securely log in using their credentials to access personalized features and app settings.



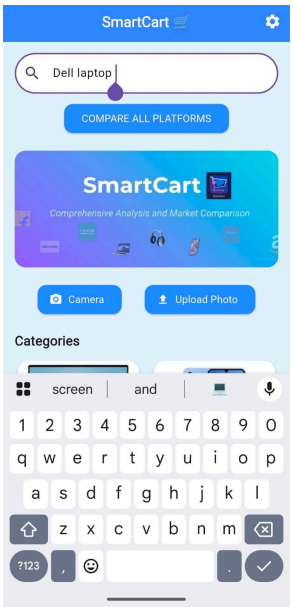
7.3 - Home screen: Product Search Interface

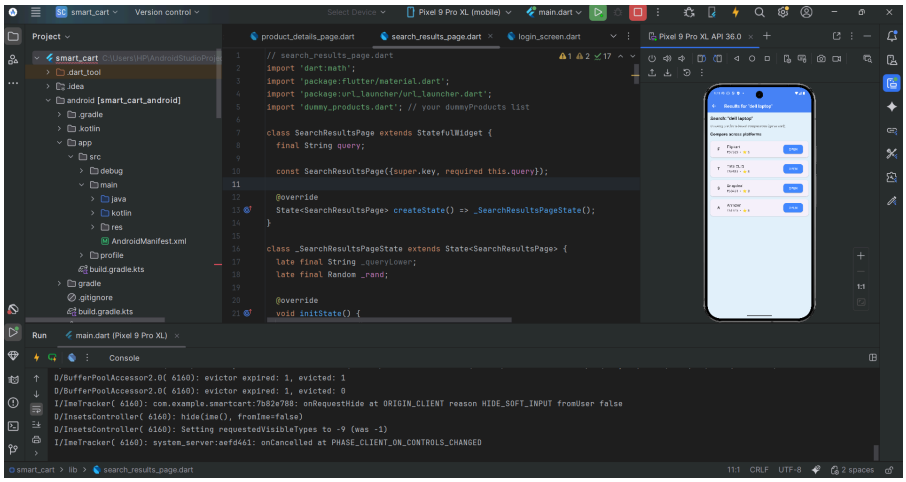
Main dashboard with search bar, image upload options, and product categories.



7.4 - Search Results Screen – Price Comparison

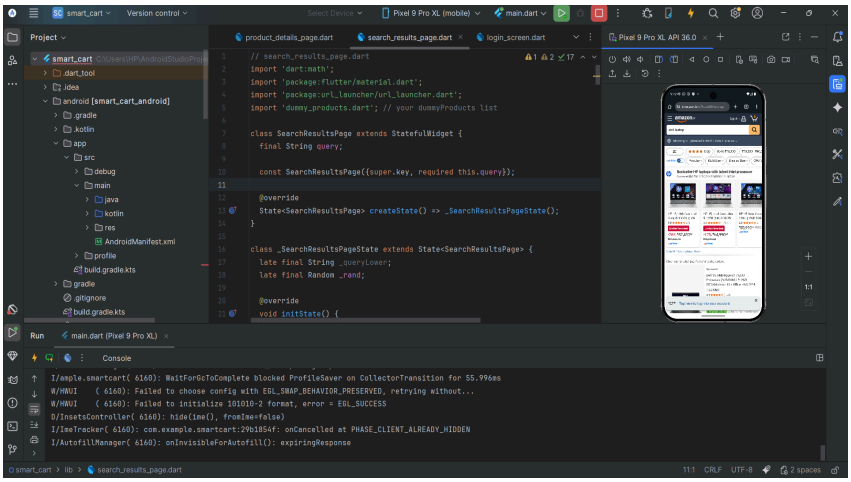
Shows product prices and ratings across multiple e-commerce platforms in ascending order.

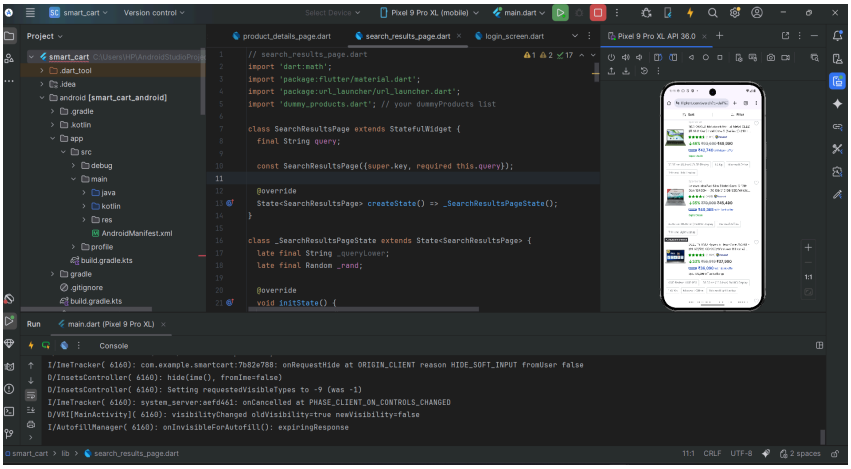




7.5 - Platform Redirection (Example: Amazon Product Page)

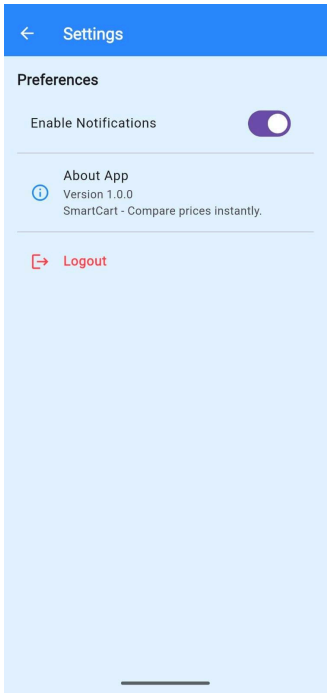
Redirects the user to the selected product's page on the respective e-commerce website.





7.6 - Settings Screen – Preferences and App Info

Contains notification settings, app version details, and logout option.



CONCLUSION

The development of the **SmartCart** application represents a significant advancement in simplifying and enhancing the online shopping experience. With the continuous rise of e-commerce platforms such as Amazon, Flipkart, and Meesho, users often struggle to identify the best deals across multiple listings. SmartCart addresses this issue by providing an intelligent price comparison system that gathers and displays product information, ratings, and prices from various platforms in a single, user-friendly interface. Developed using **Flutter** in **Android Studio**, the app supports both text-based and image-based searches, offering flexibility and convenience for all types of users. The built-in comparison engine ensures accurate sorting of results in ascending order, while platform redirection enables direct access to product pages for easy purchasing.

Through extensive testing across different devices and network conditions, SmartCart has proven to be reliable, responsive, and efficient. Each module—from data processing to result visualization—performed accurately, meeting the project's goals of saving time and improving decision-making for users. The app successfully integrates intelligent automation and mobile technology to create a modern, practical shopping tool. Moving forward, SmartCart can be expanded with real-time API integration, personalized recommendations, and user accounts, making it a scalable and future-ready solution that contributes to the evolution of smarter, AI-driven e-commerce systems.

FUTURE ENHANCEMENTS

1. Integration of Real-Time APIs: Connect directly with official APIs of e-commerce platforms like Amazon, Flipkart, and Meesho to fetch live product prices and stock availability.
2. User Account System: Enable users to create personal accounts, save favorite products, and track price changes over time.
3. Push Notifications: Notify users when the price of a saved or tracked product drops or when new offers become available.
4. Voice Search Feature: Allow users to search for products through voice commands for greater convenience and accessibility.
5. Advanced Recommendation System: Suggest similar or better products based on user preferences, search history, and reviews.
6. In-App Purchase Comparison: Introduce filters for comparing additional parameters like delivery speed, warranty, and seller reliability.
7. AI-Powered Review Analysis: Integrate sentiment analysis to summarize customer reviews and highlight product quality insights.
8. Web and iOS Versions: Expand SmartCart beyond Android by developing cross-platform compatibility for web browsers and iOS devices.
9. Offline Mode: Store recent search results temporarily so users can view comparisons even without an active internet connection.

REFERENCES

1. R. Kumar, S. Patel, "A Comparative Study of E-Commerce Price Aggregators," International Journal of Advanced Computer Science and Applications (IJACSA), 2020.

2. A. Sharma, K. Joshi, "Price Comparison Application for Online Marketplaces," International Conference on Emerging Trends in Computing and Information Sciences (ETCIS), 2021.

3. M. Singh, P. Verma, "AI-Based Product Identification Using Image Recognition," IEEE Conference on Intelligent Systems and Applications, 2022.

4. S. Roy, D. Das, "OCR-Driven Mobile Applications for Smart Shopping," International Journal of Computer Vision and Technology, 2021.

5. L. George, I. Mathew, "Design and Development of a Mobile Shopping Assistant," Proceedings of the International Conference on Computing and Information Technology, 2019.

6. K. Desai, P. Nair, "Enhancing User Experience in E-Commerce Through Mobile Apps," IEEE Transactions on Human-Computer Interaction, 2020.

7. P. Jain, R. Mehta, "Integration of Flutter and Firebase for Cross-Platform Mobile Applications," *International Journal of Software Engineering and Technology (IJSET)*, 2022.

8. Proposed System, "SmartCart – An AI-Assisted Price Comparison Application," *Student Project Report, Android Development Lab*, 2024.